

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: DRBENCHMARK | Context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Justification

The output ontology is the most severely misaligned dimension for this deployment. The STS scoring rubric is a coarse 0–5 Likert-style general semantic proximity scale [Q22, Q41] that empirically fails to distinguish legally consequential small-magnitude differences from stylistic variants — confirmed by DEFT2020-D16, DEFT2020-D17, DEFT2020-D18 (omission of safety qualifiers like 'par mesure de précaution', 'opiacé', and co-substances scored as highly similar) and CLISTER-D13 (substantially different follow-up durations scored 4.0). For NER, the entity taxonomies use UMLS Semantic Groups (QUAERO), ICD-10 chapters (DiaMED), MeSH Chapter C (DEFT-2021), and PxCorpus's 38-class prescription scheme — none of which distinguish INNs from brand names, annotate ATC codes as entities, cover excipient nomenclature, or capture EMA contraindication qualifier phrases (QUAERO-D11, QUAERO-D12, DEFT2021-D10, PXCORPUS-D19). Under the EU variation type IA/IB/II framework [WEB-6], even minor text changes have legal consequences — the benchmark scoring rubric does not encode this stricter standard. With OO weighted HIGH, this constitutes a fundamental structural validity violation.

ID	Evidence
WEB-6	Variation Type IA/IB/II classification framework establishes legal significance of small text changes — regulatory annotators would score differently than clinical/NLP annotators

Strengths

- DEFT-2021 NER provides 13 fine-grained entity types including DOSAGE, SUBSTANCE, FREQUENCY, MODE, DURATION — closest label alignment with regulatory posology entities (DEFT2021-D2, DEFT2021-D3) [Q108]
- PxCorpus 38-class NER captures detailed posology parameters (PXCORPUS-D7, PXCORPUS-D8) [Q44]
- Tokenizer-agnostic SeqEval IOB2 evaluation provides consistent scoring infrastructure [Q52, Q53]
- Two STS datasets (~2,010 annotated French biomedical pairs) provide a baseline similarity scoring infrastructure even if not regulatory-calibrated [Q41]

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories are largely irrelevant to regulatory compliance decisions. UMLS Semantic Groups [Q35, Q103] collapse INNs, excipients, and brand names under CHEM (QUAERO-D11, QUAERO-D12); ICD-10 chapters are clinical disease taxonomy; MeSH Chapter C axes are research taxonomy. None encode regulatory ontology categories.

ID	Evidence
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QUAERO-D11	Agent antithrombotique – inhibiteur direct de la thrombine , code ATC : La lépirudine ... est une hirudine recombinante — ATC code string unlabeled (tag=0); INN and endogenous enzyme both CHEM — scheme collapses regulatory-legally distinct categories
QUAERO-D12	Phosphate de sodium ... Polysorbate 80 (E433) ... Eau pour préparation injectable — Excipients and solvents all CHEM, indistinguishable from active substance

OO-2: Critical missing categories: INN-as-distinct-from-brand-name; ATC code as entity type; excipient as distinct entity; EMA contraindication qualifier phrase; population-subgroup safety qualifier; MedDRA preferred term [WEB-12, WEB-16]. The dataset analysis confirms these absences across all NER datasets (QUAERO-D11, MANTRAGSC-D8, E3C-D5, DEFT2021-D10, PXCORPUS-D19).

ID	Evidence
WEB-12	ADE Eval purpose-built English regulatory NLP achieved F1 ~0.79 — confirms difficulty of regulatory annotation alignment
WEB-16	MedDRA is the standard regulatory adverse-event terminology with French translation; not represented in DrBenchmark NER schemas
QUAERO-D11	Agent antithrombotique – inhibiteur direct de la thrombine , code ATC : La lépirudine ... est une hirudine recombinante — ATC code string unlabeled (tag=0); INN and endogenous enzyme both CHEM — scheme collapses regulatory-legally distinct categories
MANTRAGSC-D8	"Cefotaxime" tagged B-CHEM — Drug name tagged at Semantic Group level; no INN/brand/dose distinction
E3C-D5	Le patient était mis sous anti bacillaires associant Isoniazide 5mg/kg, Rifampicine 10 mg/kg, Ethambutol 25 mg/kg et Pyrazinamide 30 mg/kg — Drug INN names with per-kg dosages all tagged O; confirms drug/dosage entities absent from label scheme
DEFT2021-D10	diphenhydramine et de méthylprednisolone — Both INN drug names receive SUBSTANCE-B tag (21); no INN-specific or ATC-code annotation level
PXCORPUS-D19	kardegic 160 milligrammes 1 sachet par jour pendant 1 mois si absence d'ulcère — Contraindication phrase "si absence d'ulcère" receives no NER tag (O)

OO-3: Categories that encode non-regulatory values: STS 0–5 scale [Q22] encodes general semantic proximity, not legal equivalence. Empirically, omission of safety qualifiers and substantial numerical differences are scored as highly similar (DEFT2020-D16, DEFT2020-D17, CLISTER-D13). DEFT-2021 SUBSTANCE label conflates pharmacological and surgical treatments (DEFT2021-D9). Pharmacy company names occasionally tagged DISO (QUAERO-D18) — annotation error.

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DEFT2020-D16	En conséquence, par mesure de précaution, il convient d'éviter d'allaiter..." vs. "Il convient d'éviter d'allaiter pendant la durée du traitement. — Omission of "par mesure de précaution" scored 2–5; regulatory significance not captured
DEFT2020-D17	Ce produit peut provoquer un syndrome de sevrage opiacé..." vs. "Ce produit peut entraîner un syndrome de sevrage s'il est administré moins de 4 heures après la prise d'un stupéfiant... — Loss of "opiacé" qualifier averaged as 4.0 — regulatory specificity difference not flagged
CLISTER-D13	Le recul est de 2 ans." / "Le suivie est de 4 ans et demi. — Substantially different follow-up durations scored as highly similar
DEFT2021-D9	Par voie sous costale, il a été réalisé une néphrectomie élargie gauche, une surrénalectomie de principe associée à une splénectomie — Tag 23 (TREATMENT) applied to surgical procedures — scheme does not distinguish surgical from pharmacological treatment
QUAERO-D18	Richter Gedeon Eesti filiaal ... Polska Gedeon Richter Plc — Pharmaceutical company name annotated as DISO — likely annotation error

OO-4: Stakeholder-driven taxonomy redesign is needed for regulatory deployment: a regulatory-calibrated similarity rubric (e.g., aligned with variation type IA/IB/II thresholds [WEB-6]) and a regulatory entity ontology covering INN, ATC, excipient, contraindication qualifier, and MedDRA PT categories. No published crosswalk exists [WEB-12].

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OO-5: Severe structural validity and content validity issues: (a) STS rubric not calibrated for regulatory equivalence — confirmed full gap; (b) NER entity taxonomies do not represent regulatory ontology — confirmed across all datasets; (c) MLC

labels (ICD-10, MeSH) target clinical/research classification rather than regulatory compliance; (d) DEFT2020's mixing of encyclopedic non-medical content (DEFT2020-D11 through DEFT2020-D15) further degrades the STS signal; (e) leading model (DrBERT-FS) does not excel on STS or MLC [Q82], indicating taxonomic incoherence even within biomedical scope.

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Information Gaps

- Whether PxCorpus 38-class scheme could be reused as a partial regulatory NER ontology with relabeling

Requires Expert Verification

- Specific score thresholds in regulatory STS rubric calibrated to variation type IA/IB/II classification
- Mapping between DrBenchmark NER label spaces and EMA/ANSM annotation guidelines

Remediation

Gap 1: STS 0–5 scoring rubric not calibrated for regulatory-legal equivalence; legally consequential small-magnitude differences (omission of safety qualifiers, dose-threshold variations) scored as highly similar

Recommendation: Develop a regulatory-equivalence STS evaluation set with sentence pairs annotated by regulatory affairs specialists using a rubric aligned to EU variation type IA/IB/II classification (Type IA = no legal consequence acceptable; Type IB/II = flagged). Use existing DEFT2020 leaflet pairs as seed but re-score with regulatory authority.

Gap 2: NER label spaces do not distinguish INN from brand name, do not annotate ATC codes as entities, do not cover excipients, contraindication qualifiers, or MedDRA preferred terms

Recommendation: Construct a regulatory NER evaluation set covering INN, brand name, ATC code, excipient, posology slot fillers, contraindication qualifier, population subgroup, and MedDRA PT categories. Begin by re-annotating QUAERO EMEA and PxCorpus subsets with the new ontology, leveraging the available French MedDRA translation [WEB-11].